Zhexiang (Bruce) Li

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Available for internships: Jun 13 – Sep 17, 2027

Education

University of California, Los Angeles Expected Jun 2028
M.S. in Computer Science

University of Southern California May 2026
B.S. in Computer Science, *Magna Cum Laude*; GPA: 3.76/4.00; Viterbi Dean's List

Research Skills

Methods: multimodal learning, efficient attention, token routing, molecular property prediction
Tools: Python, PyTorch, Triton, RDKit, Chemprop, scikit-learn, C/C++, SQL, Linux

Research Experience

Health Intelligence Lab, UCLA | *Graduate Researcher* Aug 2026 – Present
Advisor: Prof. Yuzhe Yang

- Investigate multimodal glucose forecasting using CGM, insulin, meals, activity, and patient context.
- Curated a unified CGM release from **49 datasets** (**7,000+** source person records) with provenance checksums, device-clock fixes, and 14 automated release checks.
- Built patient-disjoint MetaboNet benchmarks and data audits to compare GluFormer, T1DFormer, and custom models at 30-, 60-, and 120-minute horizons.

Prof. Yan Liu's Group, USC | *Drug Discovery Research* Spring 2026
Research collaboration with a Ph.D. student in the group

- Benchmarked descriptor-based models, Chemprop, and hybrids on seven TDC datasets with scaffold splits.
- Extra Trees on RDKit descriptors had the best average AUROC (85.9), reaching **90.0** $\pm$ **6.4** on SARS-CoV-2 3CLPro; analyzed descriptor importance and SHAP attributions behind its predictions.

Selected Papers & Research Contributions

MedTokenBudget: Lesion-Preserving Token Routing 2026
Zhexiang Li (sole author). Under review.

- Developed lesion-supervised post-backbone token routing; at a 30% token budget on ISIC 2019, it achieved **80.38** $\pm$ **0.18%** accuracy versus 76.93 $\pm$ 0.03% for ToMe across three seeds, with the frozen backbone unchanged.

Mass-Conservation Token Transformers for MS/MS Retrieval 2026
Under review at AAAI 2027

- Encoded mass-conserving fragment pairs as sparse evidence tokens for spectrum retrieval; constructed mass- and overlap-based hard negatives and evaluated ranking changes after evidence removal.

Efficient Attention via Pre-Scoring 2025
Zhexiang Li, Haoyu Wang, Yutong Bao, David Woodruff. Under review. arXiv:2505.11040

- Proposed pre-scoring to prioritize informative keys before approximate attention; implemented key selection in Triton as first author, advised by Prof. David P. Woodruff (CMU).
- Integrated with HyperAttention, reduced ChatGLM perplexity from **12.0 to 9.5** at 131k-token contexts under a fixed interaction budget.

Additional Experience

Piezolab Medical | *AI Engineer Intern* Lake Forest, CA · Jun – Aug 2025

- Built an LLM-assisted pipeline to ingest, clean, and cluster **5,000+** medical-malpractice documents for quality-of-care research; developed agent workflows for vascular-care research.

University of California, Santa Cruz | *Machine Learning Tutor* Spring 2023

- Led office hours and co-designed Jupyter assignments on logistic regression, SVMs, and tree models; contributed to the coding autograder for Prof. Lise Getoor's course.